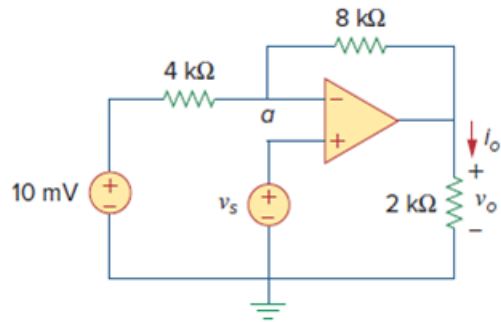


Problem

If $v_s = 0$ in the circuit of Fig. 5.41, current i_o is:

- (a) -10 mA
- (b) -2.5 mA
- (c) $10/12 \text{ mA}$
- (d) $10/14 \text{ mA}$

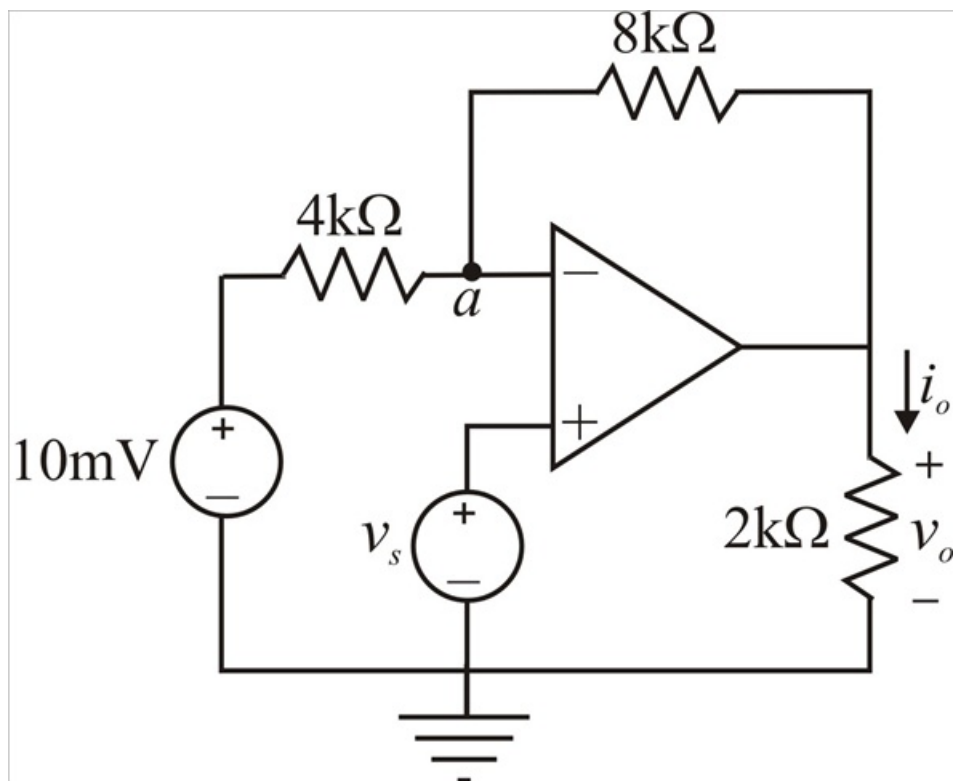
Figure 5.41:



Step-by-step solution

Step 1 of 3

Given circuit diagram



Step 2 of 3

Given $v_s = 0V$

Assume op-amp ideal one

Voltage at inverting terminal is equal to voltage at non-inverting terminal

$$v_s = v_a = 0V$$

Apply KCL at inverting terminal

$$\frac{v_a - 10 \times 10^{-3}}{4 \times 10^3} + \frac{v_a - v_o}{8 \times 10^3} = 0$$

$$\frac{0 - 10 \times 10^{-3}}{4 \times 10^3} + \frac{0 - v_o}{8 \times 10^3} = 0$$

$$\frac{v_o}{8 \times 10^3} = - \frac{10 \times 10^{-3}}{4 \times 10^3}$$

$$v_o = - \frac{10 \times 10^{-3}}{4 \times 10^3} \times 8 \times 10^3$$

$$v_o = -20 \times 10^{-3}$$

$$v_o = -20mV$$

Step 3 of 3

From the above circuit we can write

$$i_o = \frac{v_o}{2 \times 10^3}$$

$$i_o = \frac{-20 \times 10^{-3}}{2 \times 10^3}$$

$$i_o = -10 \times 10^{-6}$$

$$i_o = -10\mu A$$

Hence none of the option is correct